

Back to the ... Time Series?

Module 1 · Memory, Stationarity, and the Ghost of Spurious Regression

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Act I · What makes time series different

Time series is a different animal

A **cross-section** is a crowd photographed once. A **time series** is one subject, photographed again and again, where **order is everything**.

- Marty goes back to 1955 and shifts one punch in a parking lot — and his entire present rewrites itself.
- Shuffle a cross-section: nothing lost. Shuffle a time series: the story is gone.
- The central question of the course: **how much memory** does a series carry?
- Everything downstream is just: *how much? how do we measure it? what do we do about it?*



Past • present • future of a single process — one dashboard.

Three shapes of data

Cross-section

$$\{y_i, x_i\}, \quad i = 1, \dots, N$$

One snapshot, many subjects. **Order doesn't matter** — shuffle the rows and nothing is lost.

Time series

$$\{y_t, y_{t-k}\}, \quad t = 1, \dots, T$$

One subject, many snapshots. **Order is everything.** That dependence *is* the course.

Panel / longitudinal

$$\{y_{it}, x_{it}\}$$

Many subjects *through* time:

- **Pooled** cross-section — ignore time.
- **Repeated** cross-section — new people each period, time matters.
- **Panel** — the *same* people tracked over time.

We live in the **middle column** — but the machinery travels: a panel is time series stacked across subjects.

You never have enough time

Cross-section asymptotics lean on $N \rightarrow \infty$. Time series leans on $T \rightarrow \infty$ — and **time is stingy**.

- People on Earth: $\approx 8 \times 10^9$. Stars in view: $\approx 10^{24}$.
- Quarterly US GDP since 1947: ≈ 310 **observations**.
- The universe is only $\approx 1.4 \times 10^{10}$ years old.

...

So our estimators must work in **short, dependent** samples. That constraint shapes every choice in this course.



Depending on what you measure, you may need **A LOT** of data before the asymptotics kick in.

The master equation — one mixing board

$$y_t = \underbrace{\alpha}_{\text{level}} + \underbrace{\delta t}_{\text{trend}} + \underbrace{\sum_{j=1}^p \phi_j y_{t-j}}_{\text{AR} \cdot \text{memory}} + \underbrace{\sum_{l=1}^q \theta_l \epsilon_{t-l}}_{\text{MA}} + \underbrace{\epsilon_t}_{\text{innovation}}$$

Every model in this course is this one equation with some **faders pushed to zero**. Lecture by lecture we raise new channels.

Today only three faders are up:

$$y_t = \alpha + \phi_1 y_{t-1} + \epsilon_t$$



Model	Channels live
White noise	ϵ_t
AR(1) — today	$\alpha, \phi_1 y_{t-1}, \epsilon_t$
ARMA(p,q)	M3-M5

Two equations, never confuse them

The DGP — the truth (unobserved):

$$y_t = \alpha + \phi y_{t-1} + \epsilon_t$$

- Parameters **unhatted**: α , ϕ .
- ϵ_t — the **innovation**. The genuine surprise.

Your model — estimated (observed):

$$y_t = \hat{\alpha} + \hat{\phi} y_{t-1} + e_t$$

- Parameters **hatted**: $\hat{\alpha}$, $\hat{\phi}$.
- e_t — the **residual**. What's left over.

If your model is right, the residual e_t *behaves like* the innovation ϵ_t . When it doesn't, your model is wrong. **Every diagnostic in this course is that one sentence.**

Act II · Memory, made literal

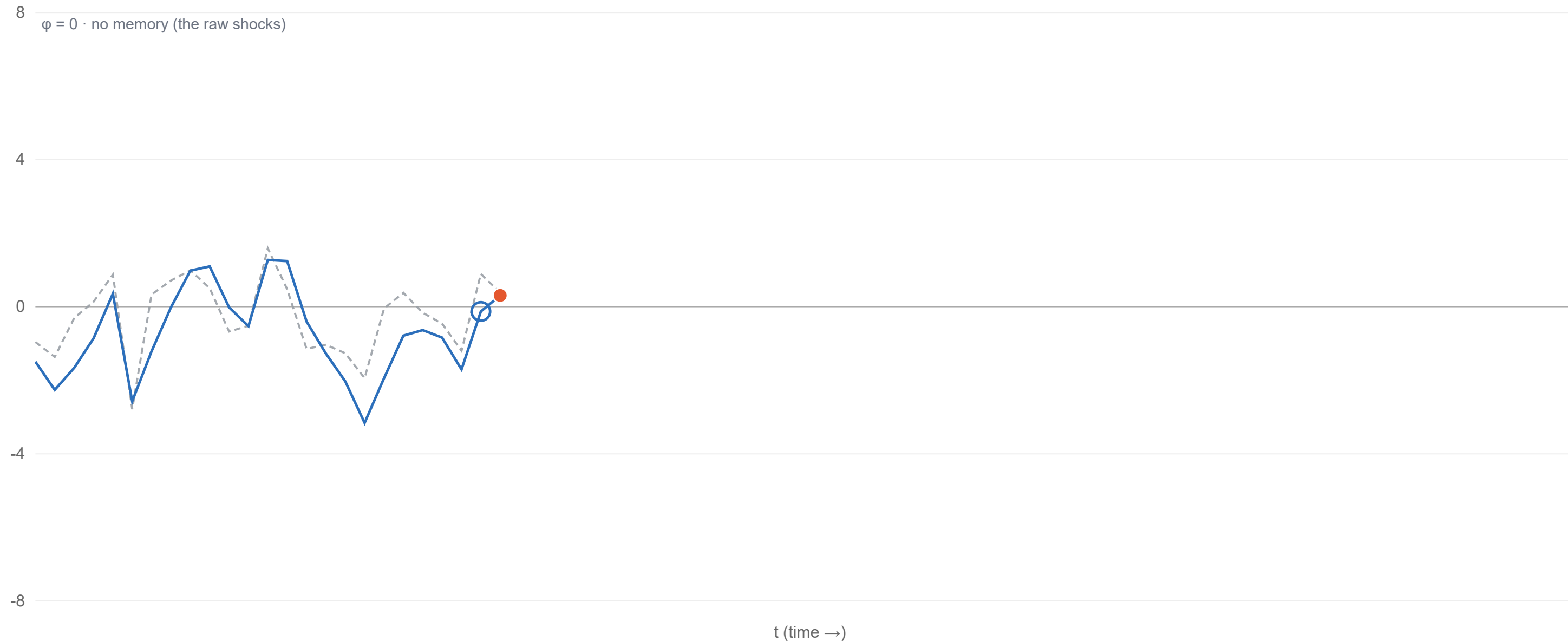
Turn off everything but the memory

$$y_t = \underbrace{\alpha}_{\text{where it lives}} + \underbrace{\phi_1}_{\text{memory}} \cdot \underbrace{y_{t-1}}_{\text{the past}} + \underbrace{\epsilon_t}_{\text{the surprise}}, \quad \epsilon_t \sim N(0, \sigma^2)$$

- ϕ_1 is the star — **how much of yesterday survives into today.**
- y_{t-1} is the series looking back at itself.
- ϵ_t is something genuinely new — a shock.

How much does the past remember?

ϕ  0.60
 || Pause
 speed  16
 Step +1
Replay



dashed grey = the same shocks with **no memory** ($\phi = 0$) — the raw surprises. blue is those shocks **remembered** at $\phi = 0.60$: the gap between the lines is the memory term $\phi \cdot y_{t-1}$, and its lean-back after each jump is the forgetting.

There's ϕ — now *feel* it.

The grey dashed line is the **same shocks with no memory** ($\phi = 0$) — the raw surprises. Drag ϕ toward **0** and the blue line collapses onto it; toward **1** and it pulls away, leaning back only slowly after each surprise. Same shocks — **only the memory changes**, and that lag is the forgetting.

Four regimes of memory

1. $\phi_1 = 0$ — **no memory**. Today owes nothing to yesterday. White noise.
2. $\phi_1 \in (-1, 1)$ — **fading memory**. The past matters, but its grip decays.
3. $|\phi_1| = 1$ — **perfect memory**. Every shock remembered forever.
4. $|\phi_1| > 1$ — **explosive**. Rare in economics; set aside.



A random walk ($\phi = 1$) is **Sheldon**: eidetic memory, every embarrassment kept forever. A stationary series is the rest of us — vivid now, faded by next month.

The for-loop *is* the equation



$y_t = \alpha + \phi \cdot y_{t-1} + \epsilon_t$
= +0.00 + (0.50)·(1.12) + (+0.51) = +1.07



Each point is assembled the same way: take ϕ of where you were (blue ring), add a fresh **shock** (orange), land at y_t . The bars are the two contributions. **Step +1** to build it one period at a time.

... which is exactly this R code

```
1 ar1_simulator <- function(n = 100, alpha = 0, phi = 0,
2                           mu = 0, sigma = 1, burn_in = 100) {
3   y <- numeric(burn_in + n)
4   for (t in 2:(burn_in + n)) {
5     y[t] <- alpha + phi * y[t-1] + rnorm(1, mu, sigma)
6   }
7   ts(y[-(1:burn_in)])      # drop burn-in
8 }
```

$$y_t = \alpha + \phi y_{t-1} + \epsilon_t$$

- The loop line **is** the equation. No translation.
- **Burn-in:** throw away the first 100 — let the process forget its arbitrary start and settle into its stationary distribution.

Production code uses `arima.sim()`; we hand-build the loop once, to feel the DGP in our fingers.

Two operators you'll see every week

Lag operator L — shift back one period:

$$L y_t = y_{t-1}, \quad L^k y_t = y_{t-k}$$

Rewrite the AR(1):

$$(1 - \phi_1 L) y_t = \alpha + \epsilon_t$$

Stationarity will live in the **roots** of that polynomial. Course convention: $\Phi(L) = 1 - \phi_1 L - \dots$ (minus signs) — flag it if you read Enders/Hamilton.

Difference operator Δ :

$$\Delta y_t = y_t - y_{t-1} = (1 - L) y_t$$

- Shows up everywhere: differencing for stationarity, $\Delta \log$ returns, ECMs.
- It's just $(1 - L)$ in disguise.

The lag operator, in code: `embed()`

The lag operator is math. `embed()` is its R twin — it stacks a series beside its own lags.

```
1 y <- c(10, 12, 15, 13, 17, 14)
2 embed(y, 3)      # cols: [y_t, y_{t-1}, y_{t-2}]
3 #               [,1] [,2] [,3]
4 # [1,]        15   12   10
5 # [2,]        13   15   12
6 # [3,]        17   13   15
7 # [4,]        14   17   13
```

- Row 1 is $(y_3, y_2, y_1) = (15, 12, 10)$ — the present laid beside its own past.
- You **lose the first $k - 1$ rows**: you can't form a lag you don't have yet.
- That block **is** the design matrix for an AR regression — regress column 1 on the rest.

Act III · The limits of memory

Stationarity: the rules don't change

A series is **weakly stationary** when:

1. $\mathbb{E}[y_t] = \mu$ for all t (constant mean)
2. $\text{Var}(y_t) = \sigma^2$ for all t (constant variance)
3. $\text{Cov}(y_t, y_{t-k}) = \gamma_k$ — depends on the **gap** k , not on *when* (stable autocovariance)

Stationarity **doesn't** mean the series sits still. It means the *rules governing its movement* don't change over time. That's what makes the past a usable guide to the future.

Strict vs. weak — which rules hold still?

Strict stationarity

$$F(y_{t_1}, \dots, y_{t_n}) = F(y_{t_1+h}, \dots, y_{t_n+h}) \quad \forall h$$

Every finite-dimensional joint distribution is unchanged by a common time shift — the *entire* distribution is frozen in time, every moment, every joint shape.

Weak (covariance) stationarity

Only the first two moments must hold still: constant mean, constant variance, and autocovariance that depends on the **gap**, not the date.

Strict $\Rightarrow$ weak, but **weak** $\nRightarrow$ **strict**. We work with **weak** stationarity throughout — it's exactly what our estimators and tests actually require.

What ϕ buys you — and where it breaks

When $|\phi_1| < 1$:

$$\mathbb{E}[y_t] = \frac{\alpha}{1 - \phi_1}, \quad \text{Var}(y_t) = \frac{\sigma^2}{1 - \phi_1^2}$$

Both are constants — no t . Conditions 1 and 2 hold.

Socratic: what happens to that variance as $\phi_1 \rightarrow 1$?

$1 - 1^2 = 0 \rightarrow$ **the variance blows up.** $|\phi_1| < 1$ isn't a convenience — it *is* the boundary of stationarity.

The autocorrelation function — memory you can plot

Autocovariance is covariance of a series with its own past:

$$\gamma_k = \mathbb{E}\left[\left(y_t - \mu\right)\left(y_{t-k} - \mu\right)\right], \qquad \rho_k = \frac{\gamma_k}{\gamma_0}.$$

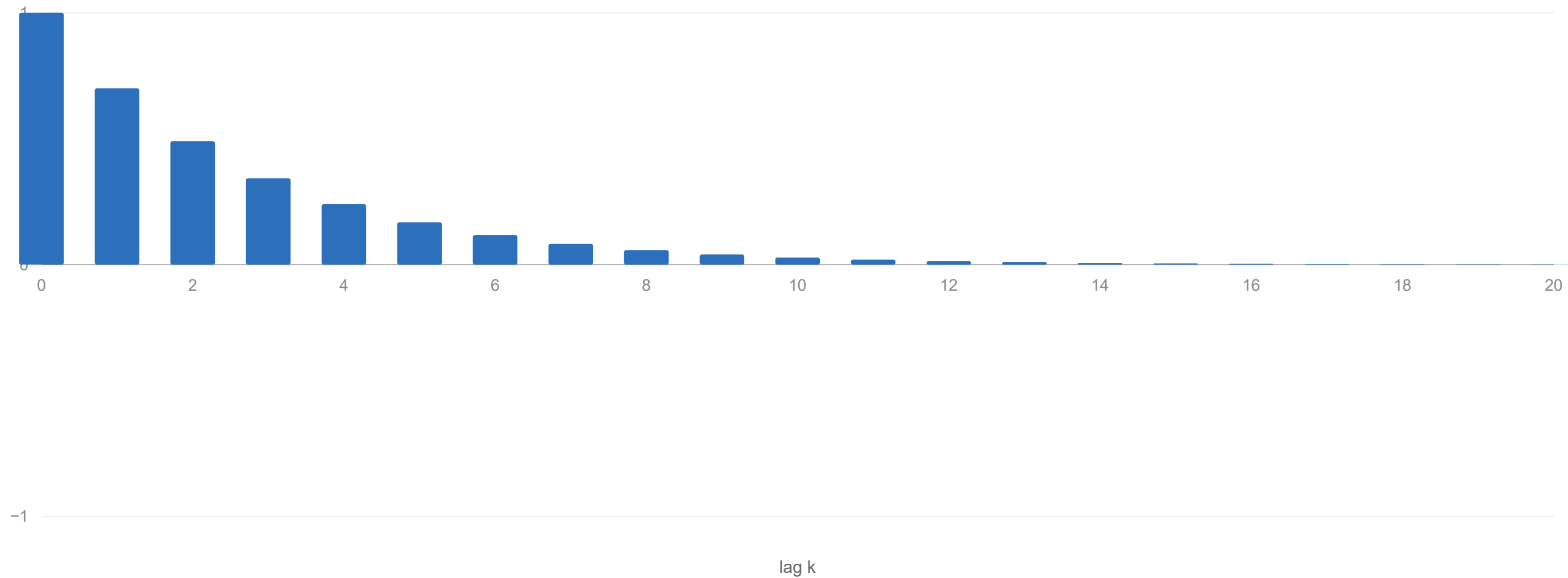
For a stationary AR(1) this collapses to something beautiful:

$$\rho_k = \phi^k \qquad \text{geometric decay.}$$

ϕ	ρ_1	ρ_2	ρ_3	ρ_{10}	memory
0.5	.50	.25	.13	.001	short
0.9	.90	.81	.73	.35	long
1.0	1	1	1	1	never decays

Watch the memory decay

ϕ 0.70



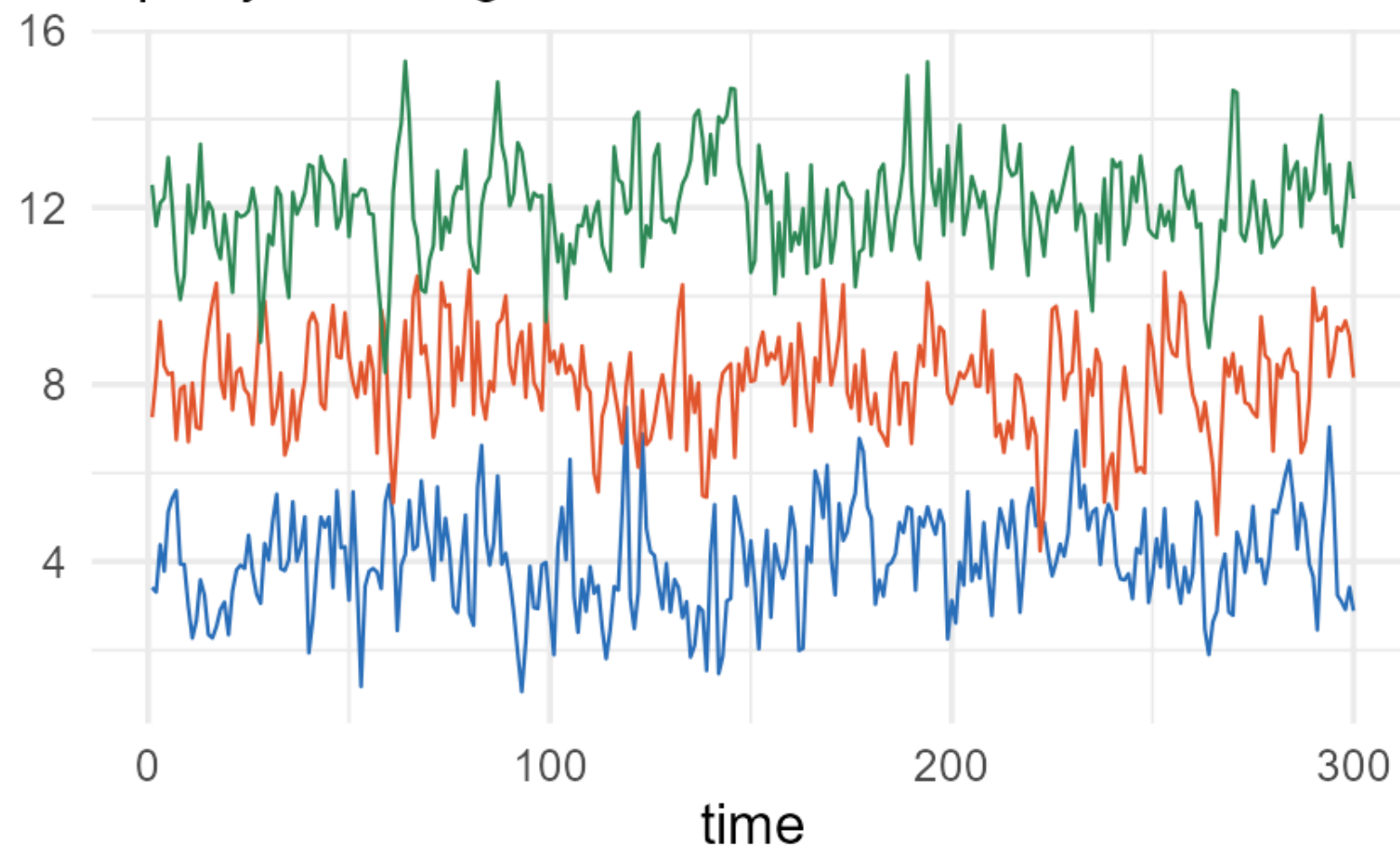
$$\rho_k = \phi^k \cdot \rho_1 = 0.70 \cdot \text{memory half-life} \approx 1.9 \text{ periods}$$

Each bar is $\rho_k = \phi^k$. Drag ϕ : toward **0** the memory collapses in a step or two; toward **1** the bars refuse to fall — the slow-decay **fingerprint of a unit root** (bars flip **orange** when $\phi < 0$, the tell-tale oscillation).

The intercept moves the level, not the memory

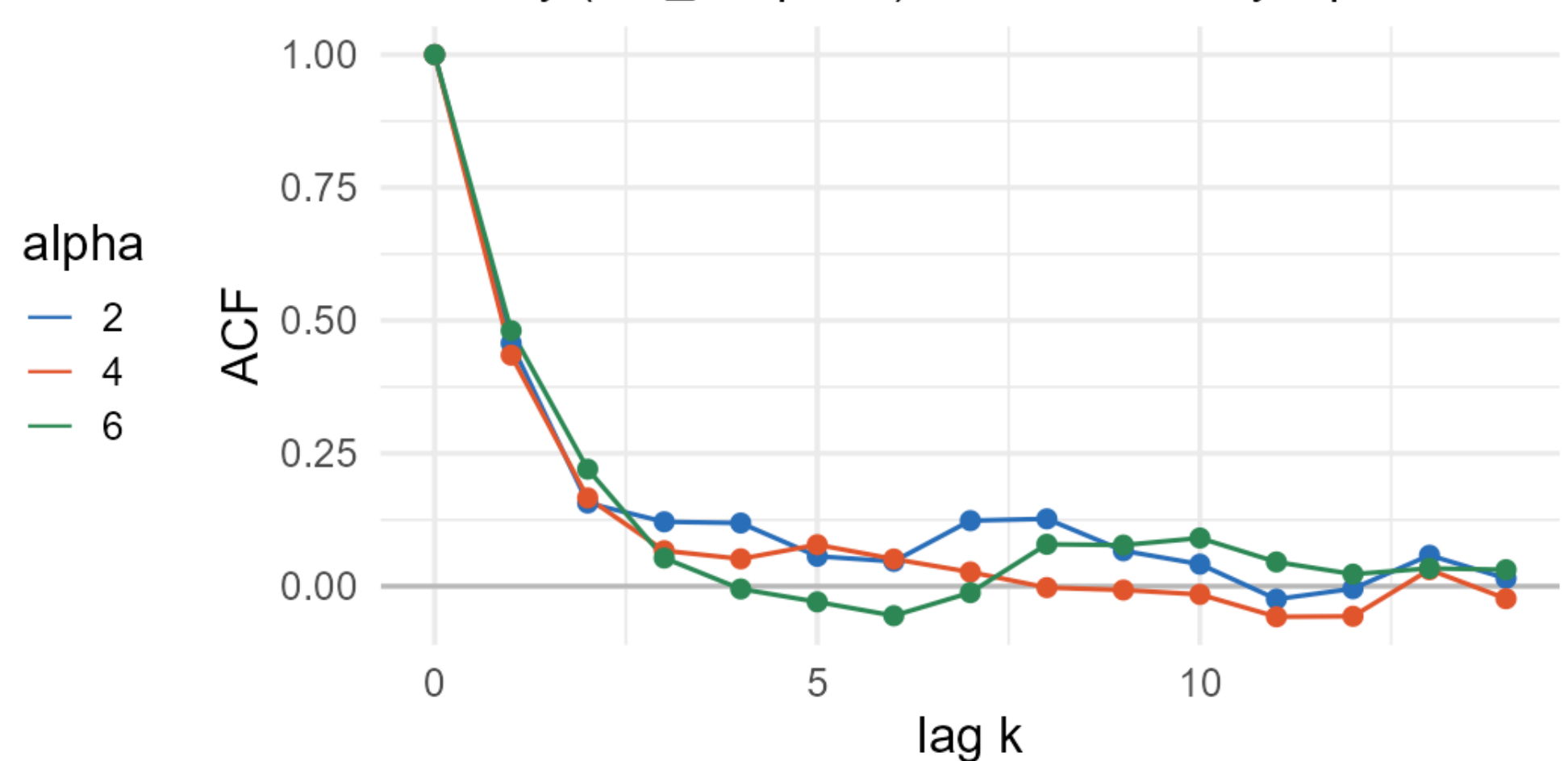
Same $\phi = 0.5$, three intercepts

α just changes where the series lives



Nearly identical ACFs

memory ($\rho_k = \phi^k$) is untouched by α



Three series, same $\phi = 0.5$, three different α . They live at different heights — yet their ACFs land right on top of each other. α sets where it lives; ϕ sets how it moves.

Shocks: temporary, or forever?

ϕ  0.70

II Pause

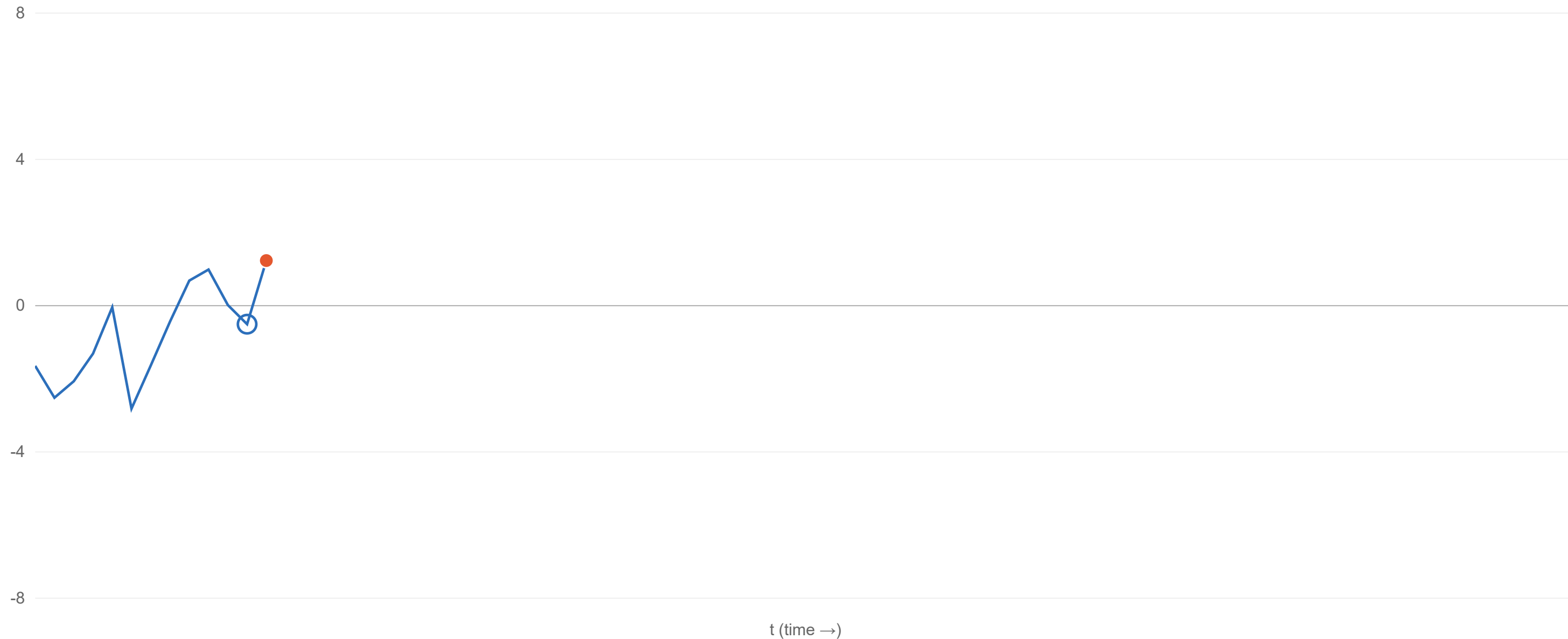
speed

 8

Step +1

Replay

 Shock



Press  Shock to inject one big surprise, then watch how long it lingers (try it at $\phi = 0.7$, then again at $\phi = 1$).

Hit  Shock. The dashed gray line is the *same series without* the shock — so the gap between them is the shock's lingering effect. At $\phi < 1$ the gap fades as ϕ^k ; at $\phi = 1$ it **never closes**. *That gap is memory, made visible* (and your first impulse-response — we'll formalize it for VARs in Module 12).

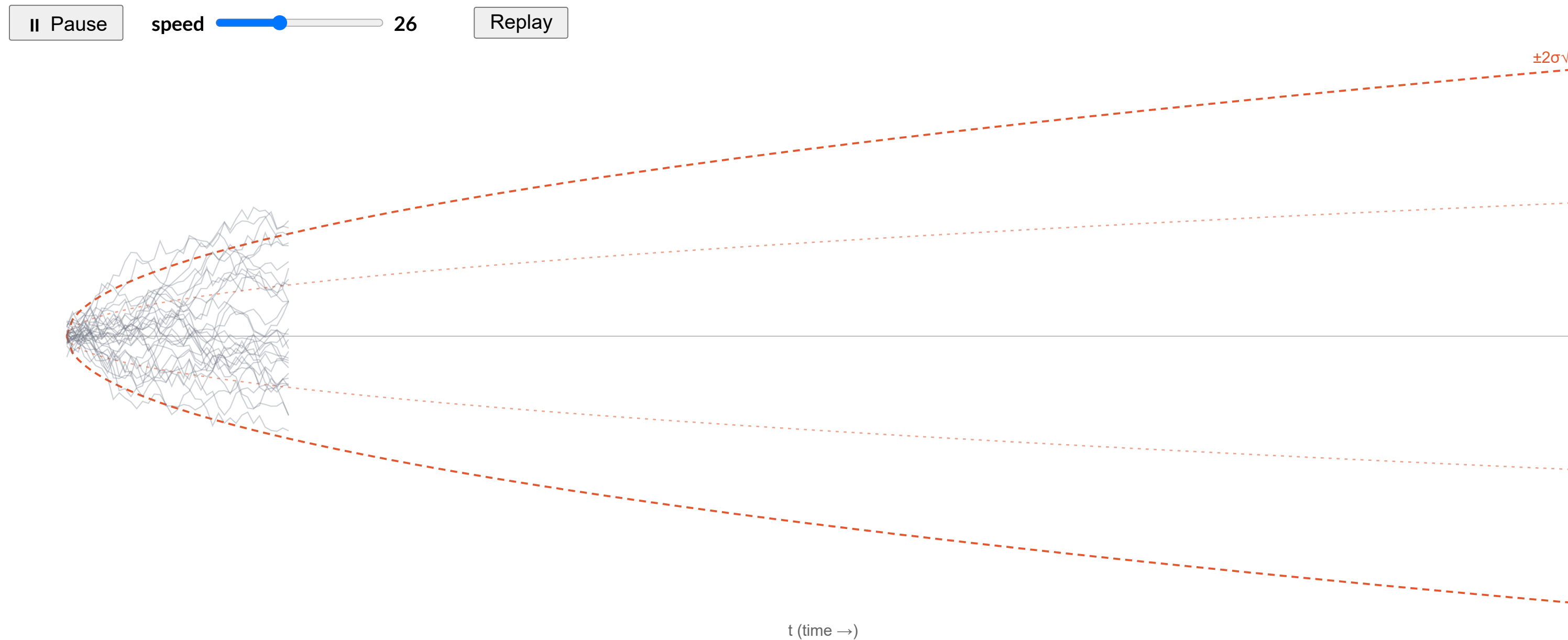
When memory is perfect: the random walk

Set $\phi = 1$:

$$y_t = y_{t-1} + \epsilon_t = y_0 + \sum_{i=1}^t \epsilon_i.$$

- The series is literally the **running total of every shock it has ever seen**.
- Nothing decays. $\text{Var} (y_t) = t \sigma^2$ — **it grows without bound**.
- Condition 2 of stationarity is dead.

The cloud that never converges



30 independent random walks · $\text{Var}(y_t) = t \cdot \sigma^2$ — the cloud widens as $\sqrt{t}$. No fixed variance to settle into → not stationary.

Thirty random walks off the same starting line, each shocked independently. The **dashed** envelope is $\pm 2\sigma\sqrt{t}$ — the spread grows without bound. A stationary series would hug a **fixed** band; this cloud **never settles**.

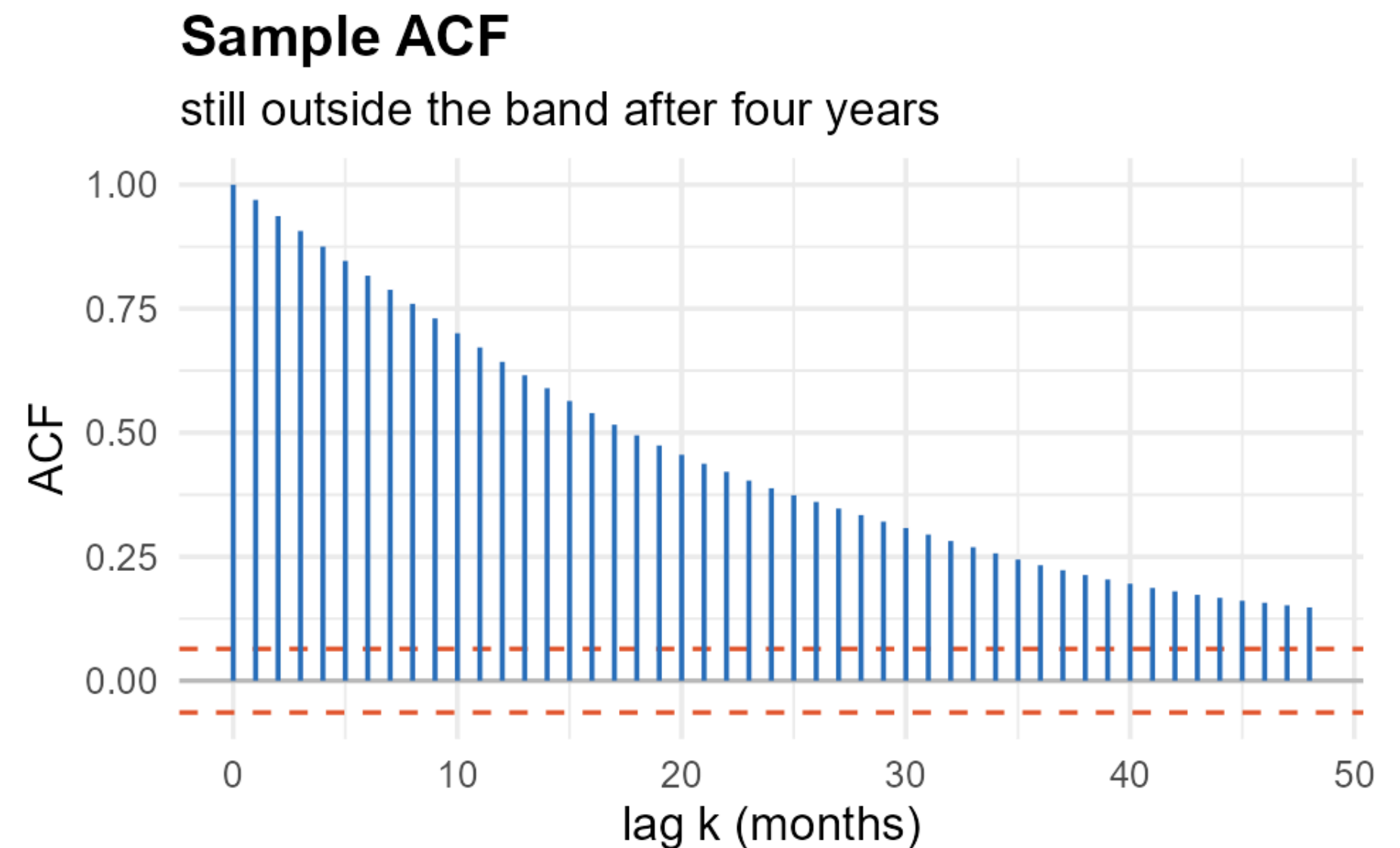
Stationary vs. random walk

Property	Stationary $ \phi < 1$	Random walk $\phi = 1$
Mean	$\frac{\alpha}{1 - \phi}$ (constant)	y_0 (fragile)
Variance	$\frac{\sigma^2}{1 - \phi^2}$ (constant)	$t \sigma^2$ (grows!)
ACF	$\rho_k = \phi^k$ (decays)	$\rho_k \approx 1$ for many lags
Shocks	temporary — fade at ϕ^k	permanent — kept forever

The random walk is Sheldon, in data form. **Eidetic memory.**

Real data: does unemployment remember?

Before the plot — commit. U.S. unemployment, monthly since 1948. Is its ACF (a) flat at zero, (b) decaying quickly to zero, or (c) barely decaying at all?



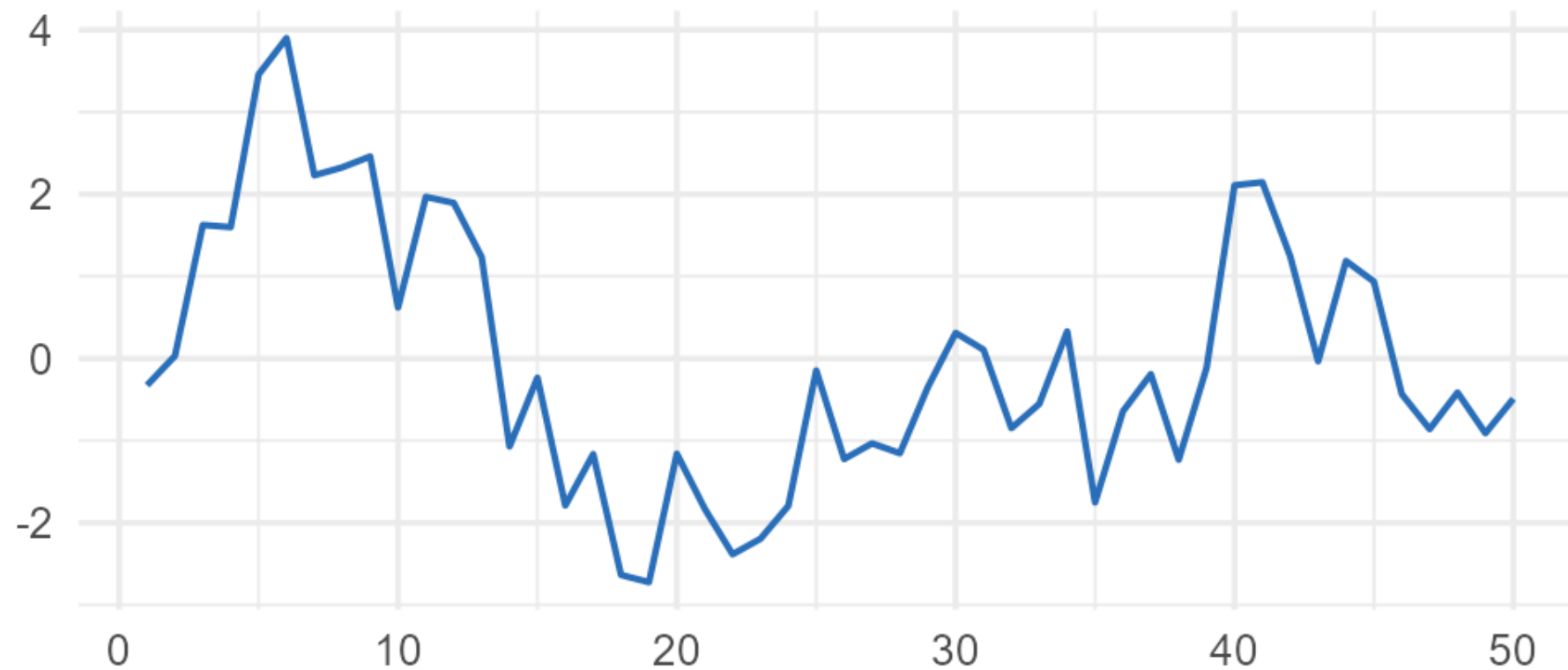
$\hat{\rho}_1 = 0.97$, and the ACF is **still outside the band at lag 48** — four years of memory. Recessions are not isolated shocks; they are sustained episodes.

But slow decay alone does not settle it. A highly persistent *stationary* process, a true unit root, and a series with structural breaks all look like this in a finite sample. We cannot yet tell which one we are holding.

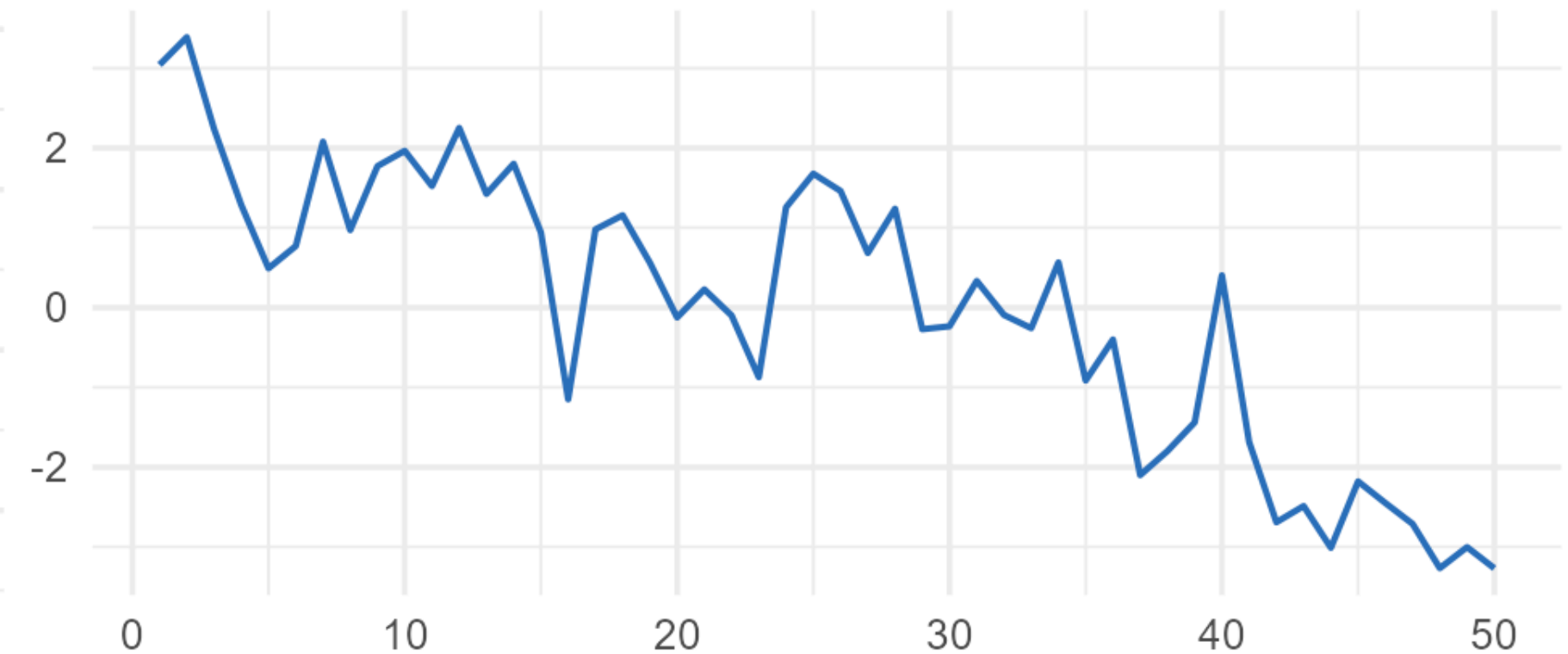
Could you tell them apart?

One is stationary ($\phi = 0.9$). One is a random walk. Which is which?

Series A



Series B



time (only 50 observations)

Series A is the stationary $\phi = 0.9$; **Series B** is the random walk. But in a **50-observation** sample, strong-but-fading memory and a true unit root look almost identical.

Eyeballing is where you **start**, not where you stop — so we need a *formal* verdict. **That test is exactly where the next module begins.**

Act IV · Why we check first

The trap: spurious regression

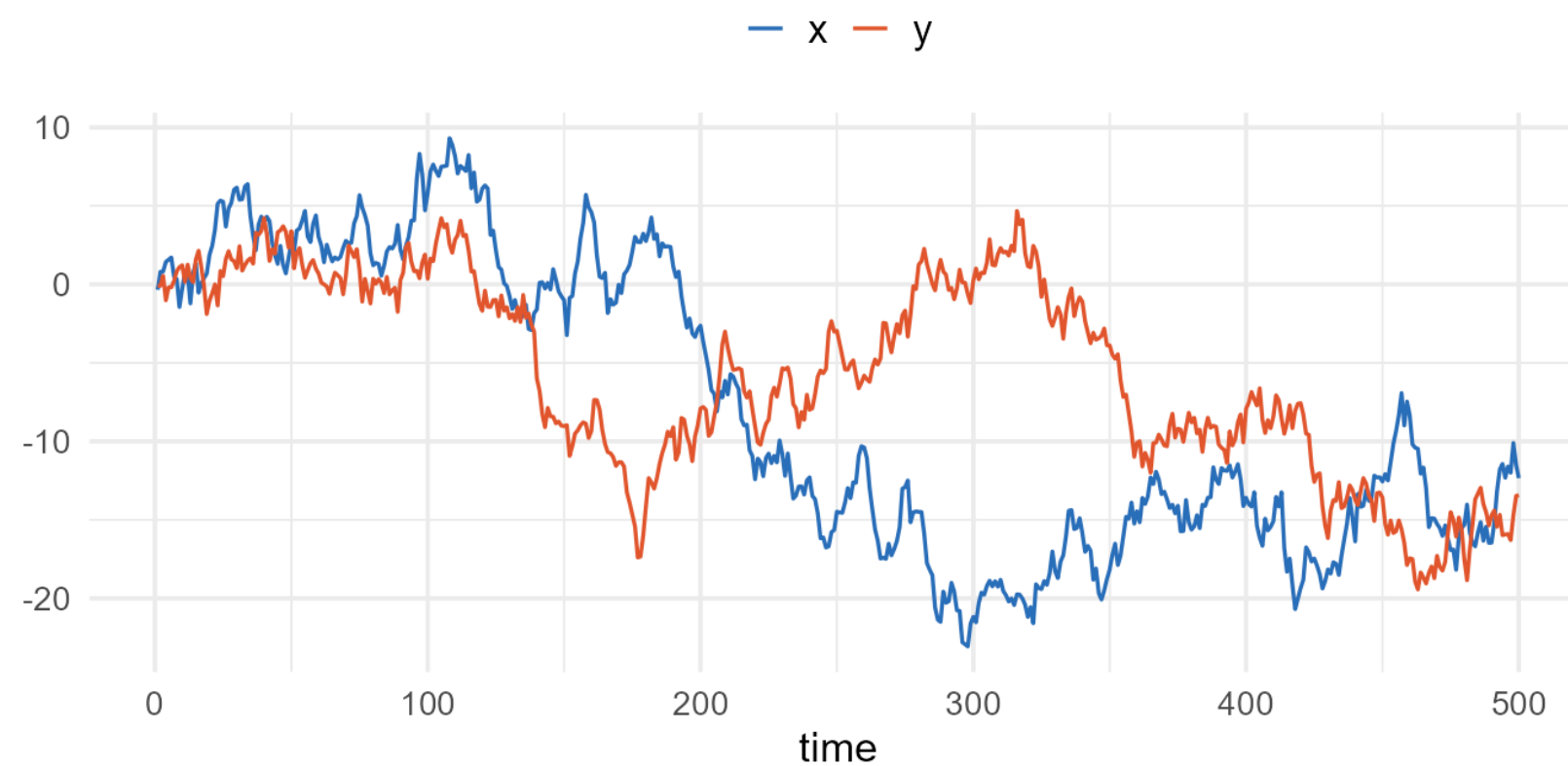
Two **non-stationary** series can look deeply related when they share *nothing*.

- This is **spurious regression** — unless the series are genuinely cointegrated (M10).
- **Granger & Newbold (1974)**: the result that made a generation of econometricians paranoid — rightly.
- OLS standard errors assume stationarity. Break that, and your t -stats lie.



Watch it happen

Two independent random walks



Regress y on x — OLS, $n = 500$:

Term	Coef.	Std. err.	t	p -value
Intercept	-3.922	0.350	-11.21	3.7×10^{-26}
x	0.211	0.029	7.30	1.1×10^{-12}

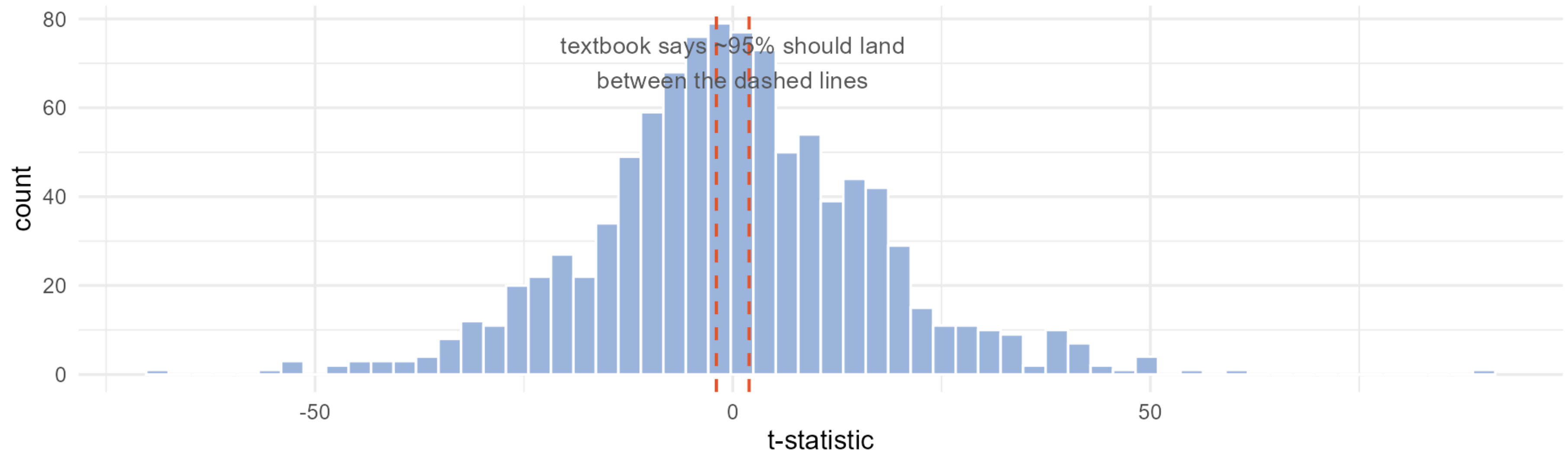
$$R^2 = 0.097 \cdot \bar{R}^2 = 0.095 \cdot F_{1,498} = 53.3 \cdot \text{resid. s.e.} = 5.89$$

A huge t , a p -value on the floor, a “real-looking” R^2 — **and there is no relationship**. Both series merely wander, and anything that wanders looks correlated with anything else that wanders over a long enough time span.

It is not a fluke

t-statistics from 1000 regressions of INDEPENDENT random walks

We reject “no relationship” 89% of the time — not the 5% the theory promises



1000 regressions of **independent** random walks, $T = 500$ each. We reject “no relationship” **89%** of the time at the 5% level. Your Type I error rate is nowhere near what you think it is — and it gets worse as T grows, not better.

The punchline

This is why stationarity is the first thing we check.

When a time-series regression hands you a beautiful R^2 , your first thought should be:

is this real — or is this Granger & Newbold?

Key takeaways

1. Time series = **temporal dependence**. Order matters; independence is gone.
2. ϕ controls memory. $|\phi| < 1$: stationary. $\phi = 1$: random walk.
3. The **ACF** measures memory: fast decay $\rightarrow$ short memory; no decay $\rightarrow$ unit root.
4. **Stationarity** = constant mean, constant variance, gap-only autocovariance.
5. Non-stationary series breed **spurious regressions** unless cointegrated — so check first.
6. ϵ_t (innovation) $\neq e_t$ (residual). This distinction runs the whole course.

Next time

- How do we **formally test** for a unit root? The Dickey–Fuller test and its augmented cousin.
- Building a case: visual inspection + ACF + formal test + rules of thumb.
- Fixing non-stationarity: differencing, and the **order of integration** $I(d)$.

See you in 1985.